

as static ranges as it is independent of the backlighting. Variants of OLED display by Samsung have logged 100 per cent of NTSC standard colour reproduction and 107% NTSC colour gamut, reproducing accurate and vibrant colours during testing

- ◆ OLEDs have a faster refresh rate and can respond to up to 200 times faster than traditional LCD displays which make it favourable for gaming and fast-moving pictures, reducing motion blur considerably.

### **Disadvantages/Cons:**

- ◆ Even though OLED production has advanced from 10% yield rate to 65% to 80%, it is still relatively low and has high wastage.
- ◆ Another milestone yet to be achieved is to bring down production costs for OLED panels along with achieving efficiency. OLED panels sell for quite a premium making it unappealing for manufacturers. Thus, not all gadgets can adopt it as it would end up increasing their manufacturing costs making the device expensive and less affordable. However, we can expect the manufacturing costs to reduce in the near future as more companies start manufacturing OLEDs.
- ◆ The lifespan of an OLED display is also a major concern. The organic molecules tend to degrade sooner as compared to the non-organic ones and earlier versions of OLED displays seem to be continuing to get worn out about four times faster than traditional LCD or LED displays.
- ◆ Another major concern in OLED panels is the nasty blue pixel. The red and the green sub-pixels have longer lifetimes up to 46,000 to 230,000 hours as compared to blue ones that last around 14,000 hours as the it deteriorates faster than the rest, thus losing out on uniform colour reproduction across the panel.

As a way around to this issue, Samsung has been following a double-blue pixel approach wherein a blue pixel, twice as large than the green and the red one is added in the arrangement to combat the faster decay of the blue sub pixel compound. LG, on the other hand, uses the WRGB arrangement to combat this issue, but a more reliable and efficient fix is expected.

### **Types of OLEDs**

Currently, there are total six types of OLED, out of which Active matrix OLED and Passive matrix OLED are the primary means of addressing pixels. Read on to know about them: